The documentation and process conversion measures necessary to comply with this amendment shall be completed by 2 Oct 91

INCH-POUND

MIL-S-19500/565A
AMENDMENT 2
2 July 1991
SUPERSEDING
AMENDMENT 1
12 May 1988

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT
TRANSISTOR, P-CHANNEL, SILICON,
TYPES 2N6895, 2N6896, 2N6897 AND 2N6898
JANTX, JANTXV AND JANS

This amendment forms a part of MIL-S-19500/565A, dated 7 October 1987, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.3, footnote 2/, line 2: Delete "(T-25)" and substitute "(T_C -25)".

PAGE 7

4.3.1, second line: Delete "-0°C" and substitute "-5°C".

4.5.3, delete and substitute the following:

"4.5.3 Thermal response (V_{SD} measurements). The ΔV_{SD} measurements shall be performed in accordance with method 3161 of MIL-STD-750. The ΔV_{SD} conditions (I_H and V_H) and maximum V_{SD} limit shall be derived by each vendor from the thermal response curves (see figure 4). The chosen ΔV_{SD} measurement and conditions for each device in the qualification lot shall be submitted in the qualification report. The chosen ΔV_{SD} shall be considered final after the manufacturer has had the opportunity to test five consecutive lots."

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* The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Pages replaced
9	9
10	10
11	11
12	12

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PAGE 13

TABLE IIa, subgroup 4, Intermittent operation life, conditions column: Delete "see 4.3.1"; "V_{DS} = -10 V dc; P_T = .6 W at"; and "V_{DS} = -20 V dc; P_T = 4 W at".

PAGE 14

TABLE IIa, subgroup 5, Accelerated steady-state operation life, conditions column: Delete "T_J = 200°C" and substitute "(except) T_J = 200°C, +20°C, -0°C".

PAGE 15

TABLE IIb, subgroup 3, Intermittent operation life (LTPD), conditions column: Delete "A" and substitute "The heating".

PAGE 16

TABLE III, subgroup 2: Delete "Visual and mechanical evaluation" and Method "2071".

TABLE III, subgroup 6, Intermittent operation life (LTPD), conditions column: Delete "A" and substitute "The heating".

PAGE 18

TABLE IV, step 3, Gate current, conditions column: Delete "V_{GS} = -20 V dc" and substitute "V_{GS} = +20 V dc and -20 V dc".

TABLE IV, step 4, Drain current, max limits column: Delete "-25" and substitute "-50".

TABLE IV, footnote 1/: Delete in its entirety.

PAGE 20

FIGURE 4, Transient thermal response, note 2: Delete "$R_{\theta JC}$ = 1.67°C/W" and substitute "$R_{\theta JC}$ = 2.083°C/W".

CONCLUDING MATERIAL, Custodians, Review activities, and User activities: Delete and substitute as printed herein.

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 13, 15

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1233)

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TABLE I. Group A inspection.

Inspection	MIL-STD-750 Method	MIL-STD-750 Conditions	LTPD 1/ JANS	LTPD 1/ JANTX, JANTXV	Symbol	Limits Min	Limits Max	Unit
Subgroup 1								
Visual and mechanical inspection	2071							
Subgroup 2								
Breakdown voltage, drain to source (threshold)	3407	I_D = -1.0 mA dc, bias condition C, V_{GS} = 0 V			$V_{(BR)DSS}$	-100		V dc
Gate to source voltage	3403	$V_{DS} \geq V_{GS}$, I_D = -0.25 mA dc			$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate current	3411	V_{GS} = +20 V dc and -20 V dc, bias condition C, V_{DS} = 0 V			I_{GSS1}		±100	nA
Drain current	3413	V_{DS} = -80 V dc, bias condition C, V_{GS} = 0 V			I_{DSS1}		-1.0	µA
Static drain to source on-state resistance	3421	V_{GS} = -10 V dc, pulsed (see 4.5.1), condition A			$r_{DS(on)1}$			
2N6895		I_D = -0.74 A dc					3.65	Ohm
2N6896		I_D = -3.8 A dc					0.6	Ohm
2N6897		I_D = -7.6 A dc					0.3	Ohm
2N6898		I_D = -15.8 A dc					0.2	Ohm
Drain to source on-state voltage	3405	V_{GS} = -10 V dc, pulsed (see 4.5.1), condition A			$V_{DS(on)1}$			
2N6895		I_D = -1.16 A dc					-6.0	V
2N6896		I_D = -6.0 A dc					-6.0	V
2N6897		I_D = -12.0 A dc					-4.8	V
2N6898		I_D = -25.0 A dc					-6.0	V
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1), V_{GS} = 0 V			V_{SD}			
2N6895		I_S = -1.16 A dc				-0.8	-1.6	V
2N6896		I_S = -6.0 A dc				-0.8	-1.6	V
2N6897		I_S = -12.0 A dc				-0.8	-1.6	V
2N6898		I_S = -25.0 A dc				-0.8	-1.6	V
Forward transconductance	3475	I_D = Rated I_{D2} (see 1.3), pulsed (see 4.5.1)			g_{FS}			
2N6895						0.2	.8	S
2N6896						1.0	4.0	S
2N6897						2.0	8.0	S
2N6898						4.0	16.0	S

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 3								
High temperature operation:		$T_C = T_J = +125°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = -0.25$ mA dc			$V_{GS(th)2}$	-1.0		V dc
Gate reverse current	3411	$V_{GS} = +20$ V dc and -20 V dc, bias condition C, $V_{DS} = 0$ V			I_{GSS2}		±200	nA dc
Drain current	3413	$V_{DS} = -80$ V dc, bias condition C, $V_{GS} = 0$			I_{DSS2}		-50	µA dc
Static drain to source on-state resistance	3421	$V_{GS} = -10$ V dc, pulsed (see 4.5.1)			$r_{DS(on)2}$			
2N6895		$I_D = -0.74$ A dc					5.66	Ohm
2N6896		$I_D = -3.8$ A dc					0.96	Ohm
2N6897		$I_D = -7.6$ A dc					0.465	Ohm
2N6898		$I_D = -15.8$ A dc					0.31	Ohm
Low temperature operation:		$T_C = T_J = -55°C$						
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, $I_D = -.25$ mA dc			$V_{GS(th)3}$		-5.0	V dc
Subgroup 4								
Switching time test	3472	I_D = Rated I_{D2} (see 1.3), $V_{GS} = 10$ V dc, Gate drive impedance = 7.5Ω						
Turn-on delay time					$t_{d(on)}$			
2N6895		$V_{DD} = -50$ V dc					25	ns
2N6896		$V_{DD} = -50$ V dc					60	ns
2N6897		$V_{DD} = -50$ V dc					60	ns
2N6898		$V_{DD} = -50$ V dc					50	ns
Rise time					t_r			
2N6895		$V_{DD} = -50$ V dc					45	ns
2N6896		$V_{DD} = -50$ V dc					100	ns
2N6897		$V_{DD} = -50$ V dc					175	ns
2N6898		$V_{DD} = -50$ V dc					250	ns
Turn-off delay time					$t_{d(off)}$			
2N6895		$V_{DD} = -50$ V dc					45	ns
2N6896		$V_{DD} = -50$ V dc					150	ns
2N6897		$V_{DD} = -50$ V dc					275	ns
2N6898		$V_{DD} = -50$ V dc					400	ns

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 4 - continued								
Fall time					t_f			
2N6895		V_{DD} = -50 V dc					50	ns
2N6896		V_{DD} = -50 V dc					100	ns
2N6897		V_{DD} = -50 V dc					175	ns
2N6898		V_{DD} = -50 V dc					250	ns
Subgroup 5								
Safe operating area	3474	See figure 3, V_{DS} = 80% of rated V_{DS} and $V_{DS} \leq 200$ V maximum						
High voltage dc SOA		t_p = 1 s						
Electrical measurements		See table IV, steps 1, 2, 3, 4, 5, 6, and 7						
Subgroup 6								
Not applicable								
Subgroup 7								
Gate charge	3471	Condition A and B						
Test 1								
On-state gate charge					$Q_{g(on)}$			nC
2N6895						2.2	4.7	
2N6896						13	24	
2N6897						31	58	
2N6898						50	100	

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-750		LTPD 1/		Symbol	Limits		Unit
	Method	Conditions	JANS	JANTX, JANTXV		Min	Max	
Subgroup 7 - continued								
Test 2								
Gate to source charge					Q_{gs}	0.4 1.1	1.2 5.5	nC
2N6895						3	13	
2N6896						6	25	
2N6897								
2N6898								
Test 3								
Gate to drain charge					Q_{gd}			nC
2N6895						0.9	2.9	
2N6896						5.5	14.5	
2N6897						14	36	
2N6898						26	69	
Reverse recovery time	3473	$V_{DD} \leq 30$ V, $di/dt = 100$ A/µs			t_{rr}			
2N6895		$I_F = 4$ A					340	ns
2N6896		$I_F = 4$ A					375	ns
2N6897		$I_F = 4$ A					500	ns
2N6898		$I_F = 4$ A					750	ns

1/ For sampling plan, see MIL-S-19500.

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